

Information for first and second responders

Emergency response guide



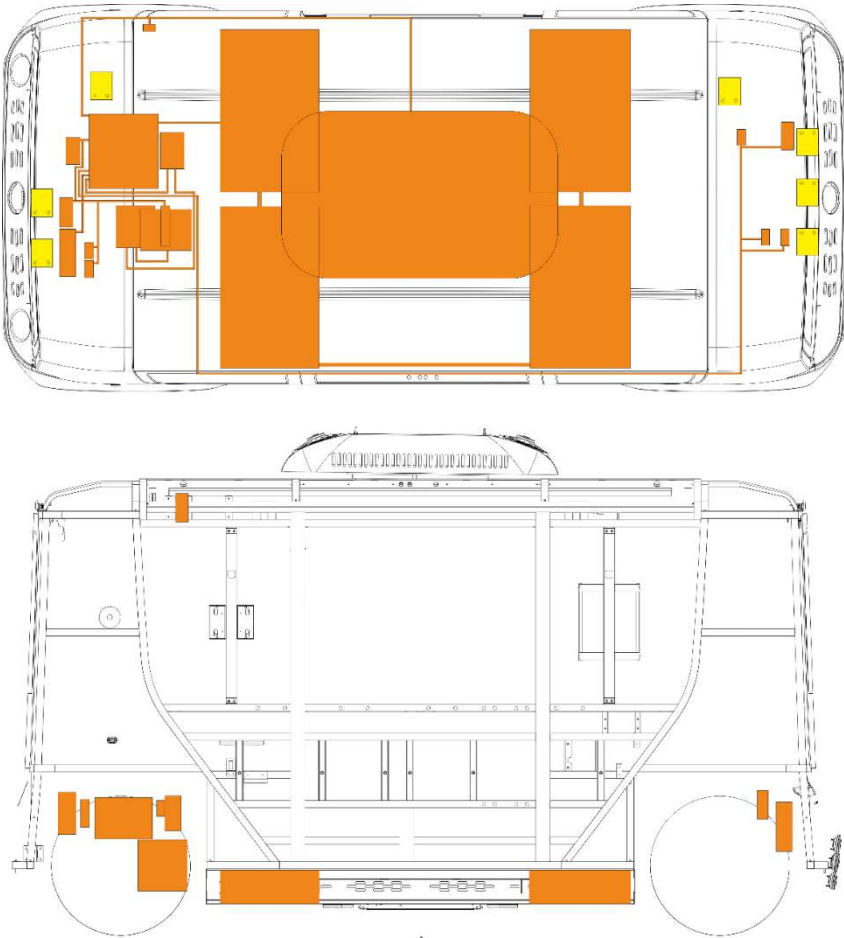
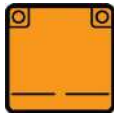
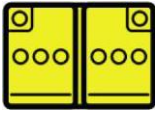
Ohmio Lift

100% Electric vehicle



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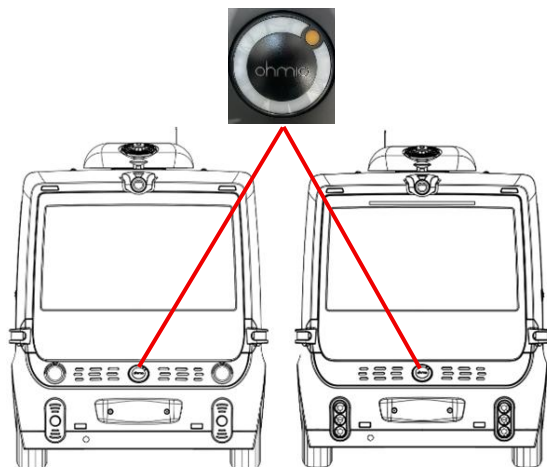
		Ohmio Lift From 2020-present			
					
					
		High voltage battery and components			
					
				High voltage power cable	



LACK OF ENGINE NOISE DOES NOT MEAN THE VEHICLE IS OFF. SILENT MOVEMENT OR INSTANT RESTART CAPABILITY EXISTS UNTIL THE VEHICLE IS FULLY SHUT DOWN. WEAR APPROPRIATE PPE

Badging

The lettering “Ohmio” can be found both front and back located on the central black flap as well as on the wheels



LIDARS

One of the primary methods of obstacle detection and pathfinding is the LIDARS. There are 3 on each side of the vehicle to make a total of 6. These LIDARS are mirrored front and back so only one side is shown here.



VIN

The VIN is located under the seat on the front side of the vehicle. Lifting the seat and the panel underneath will reveal the VIN to the right side. There is also a secondary VIN located in the right front wheel well stamped into the struts. The Ohmio Lift can be identified with the “L” in the 4th alphanumeric position.

VIN under seat



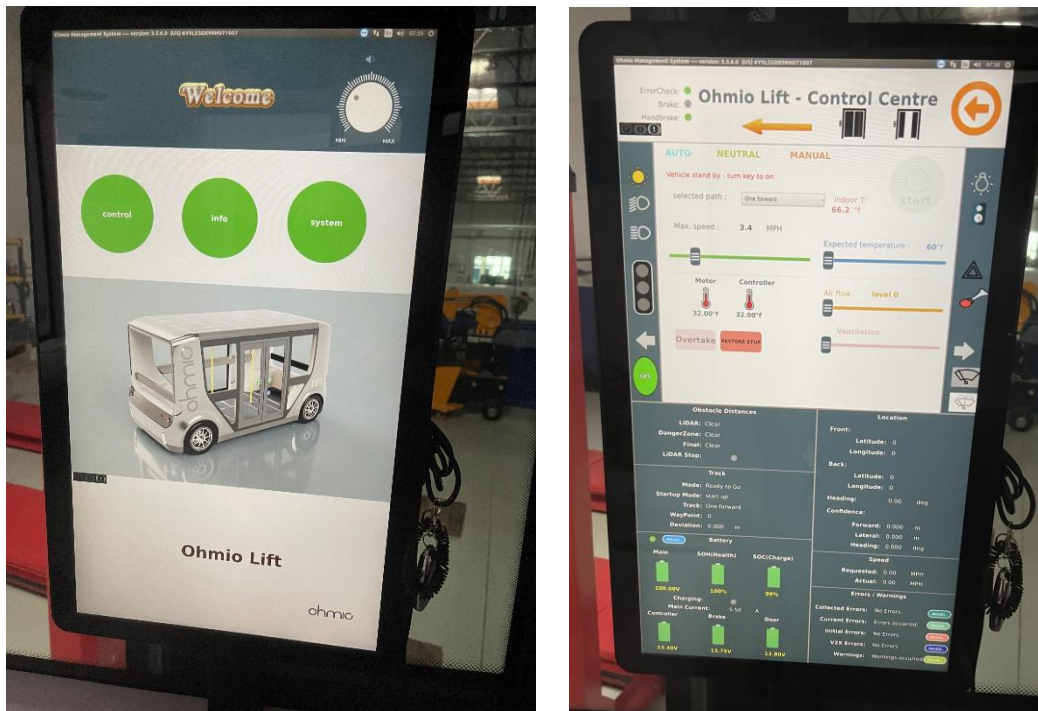
VIN in wheel well



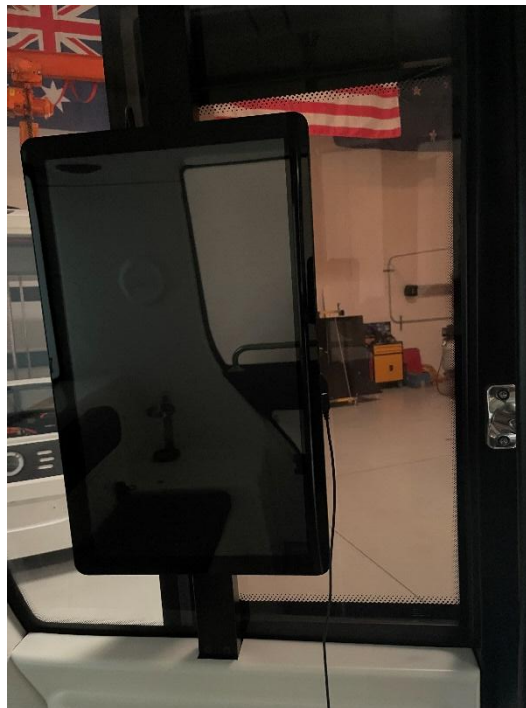
Touchscreen

The Ohmio Lift makes use of a 23-inch (58cm) touchscreen to operate the autonomous vehicle that is mounted in a portrait orientation and is located on the door side of the vehicle to right as you enter the vehicle. There are no controls as would be in a vehicle that is driven by a person. Should the main battery be disabled or otherwise inoperative this touchscreen will not be operational.

Touch screen under normal operation



Touch screen – unpowered



Keys

The keys for the vehicles are located on the outside of the vehicle under the central black “Ohmio” flap on the Rear

Model: Ohmio lift

Document number: Ohmio-ERG-002

Version: 1

IMMOBILIZATION

Key

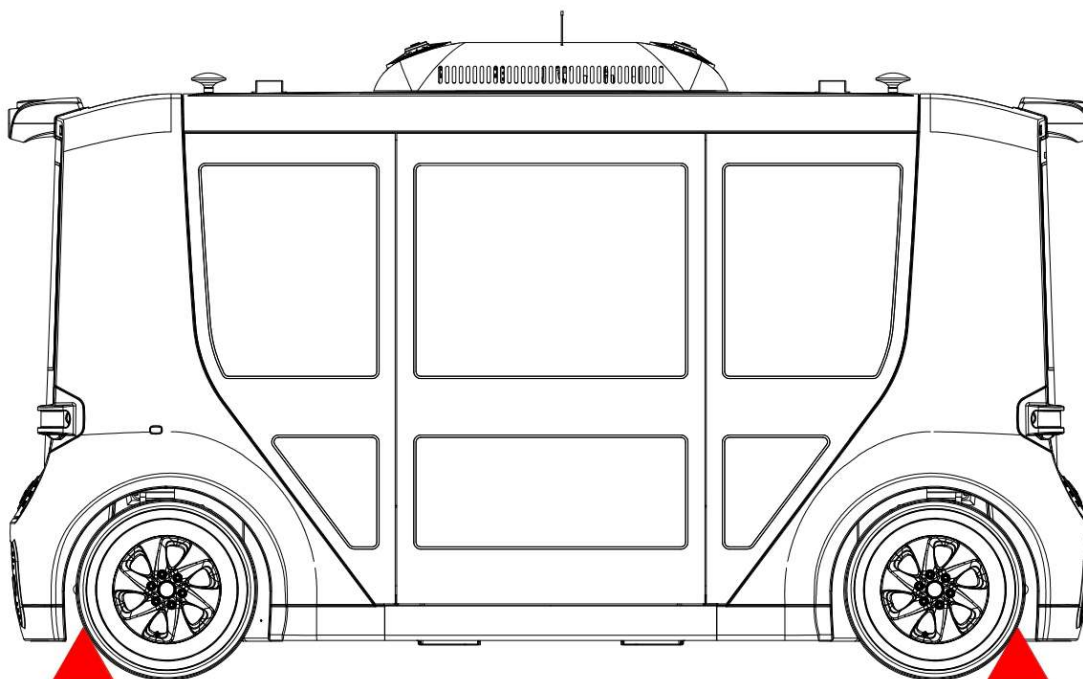
The key for the vehicle is located outside the cabin under the rear “Ohmio” flap. Turning this key to the off position will disengage power to the motor and begin the shutdown sequence of the vehicle. This process can take up to 2 minutes during which time the system will engage the parking brakes.

Automatic handbrake

After 30 seconds of no movement the autonomous system will automatically engage the parking brake. However, after a crash incident it is possible that the autonomous system will malfunction. There are several steps that can be taken to reduce the chance of vehicle movement

Chock wheels

The Ohmio Lift has an integrated AVAS (Acoustic Vehicle Alerting System) but never assume that it is deactivated. The vehicle should not move unless commended to do so but after a crash incident there is no guarantee that something hasn't happened to the control system to make the vehicle suddenly move. To avoid this situation, make sure to place wheel chocks in front of and behind the wheels



System override

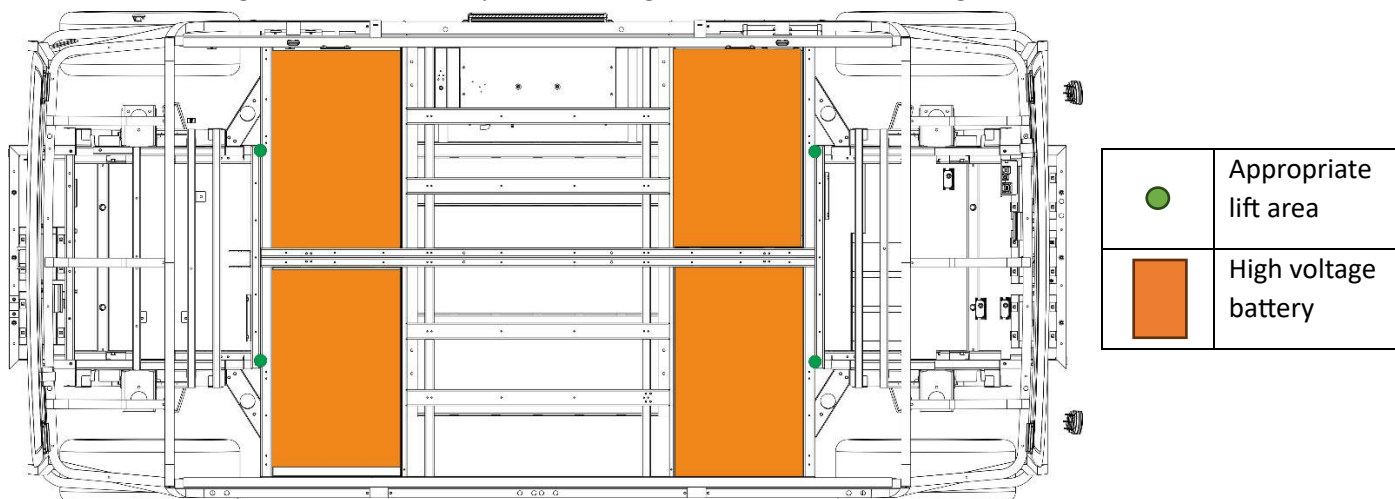
Inside the vehicle there are 2 emergency stop buttons. As well as a remote emergency stop. By pressing any one of these buttons the vehicle will enter a state of emergency, disengage the motor power, and engage the both the hydraulic and parking brakes

STABILIZATION

Due to the large sides of the Ohmio Lift it will be naturally stable when tipped to its side, so additional stabilization is unlikely to be required

LIFTING

The main battery is located within the floor of the Ohmio Lift and covers areas to both the front and rear of the vehicle. When lifting the Ohmio Lift, only use the designated lift areas shown in green.



Do not use the main battery to lift the vehicle.
Take care to not damage the main battery while lifting

VEHICLE KEY

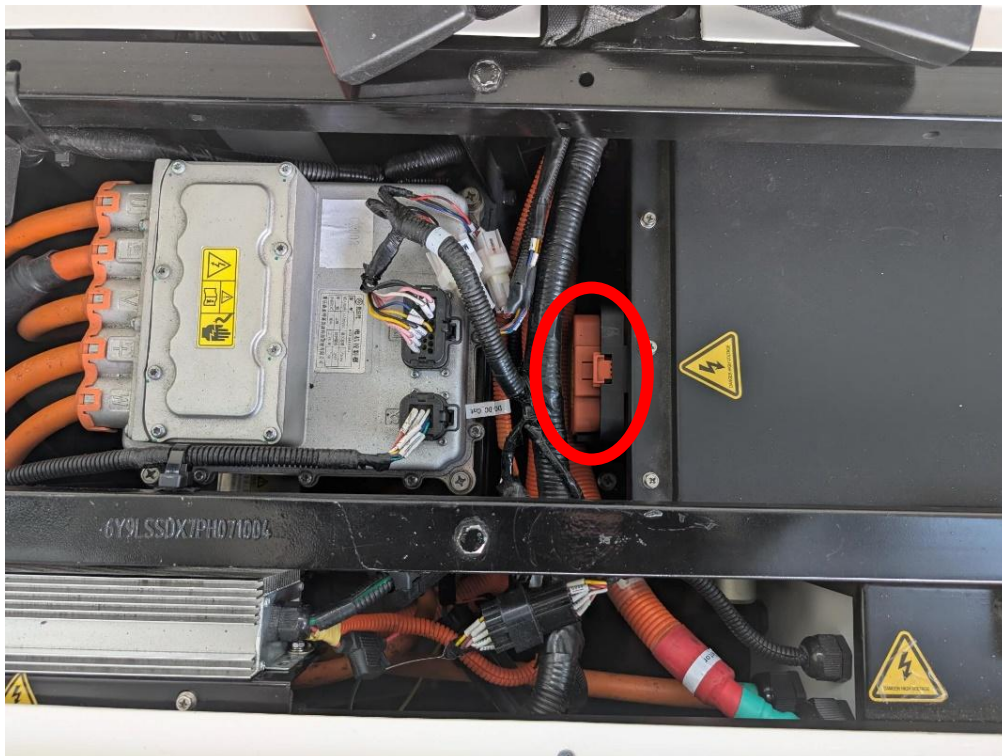
The key for the Ohmio Lift is located outside of the vehicle cabin so can be accessed without having to enter the vehicle. By turning the key to the off position, it will cause the vehicle to begin its shutdown sequence. Part of which is to isolate the main battery from the rest of the vehicles electrical system. This process does take around 2 minutes to complete.

The key is located behind the “Ohmio” flap at the rear of the vehicle. To access the key, press the flap in and the catch will release. The key can then be turned anticlockwise to the off position



MAIN ISOLATION KEY

There is an isolation key located within the cabin of the vehicle under the front seat. To access the isolation key first remove the seat which is held in place by Velcro, remove the cover that is under the seat, then remove the isolation key. The front seat is located to the right as you enter the cabin of the vehicle.



4. Access to the occupants

NOTE: The door requires power from the main battery to operate. As such it might not operate after a collision. If the button is illuminated green, then the door has power.

NOTE: The Ohmio Lift does not have airbags or seatbelt pretensioners

OPENING THE DOOR WITHOUT POWER

To open the door without power there is mechanical release on both the inside and outside of the vehicle. To operate the release first open the cover by pressing the red release catch, then turn the release as far clockwise as is possible. The door should slightly crack open and can then be completely opened by hand by pushing the doors apart. The release is located to the rear of the vehicle on the door side.

Internal door release



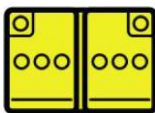
External door release

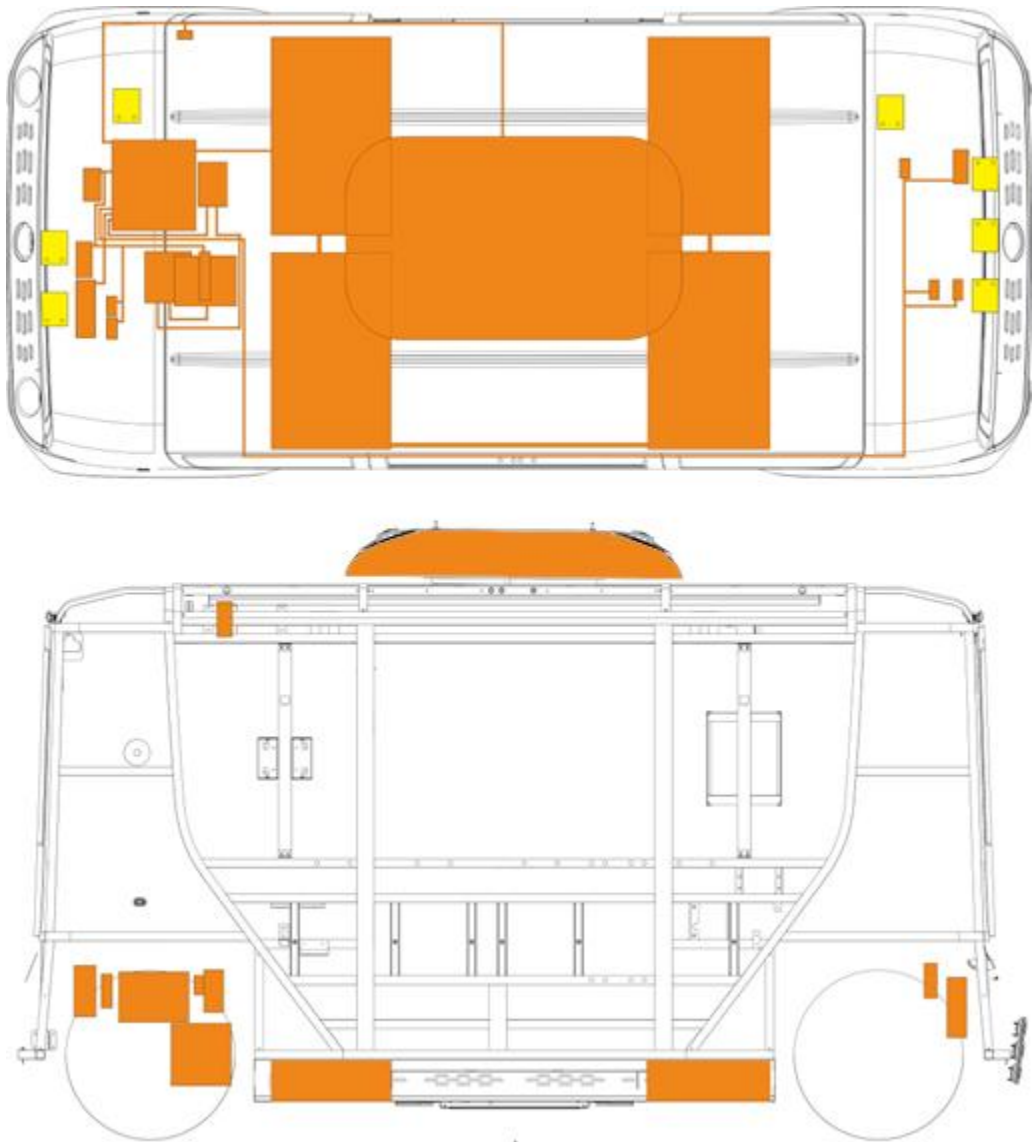


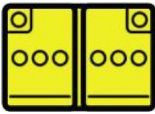
OPENING THE DOOR WITH POWER

If the door button is illuminated green, then tapping the button will open the doors. If after pressing the button the door doesn't open or open completely then follow the instructions as if the door has no power.

5. Stored energy / liquids / gases / solids

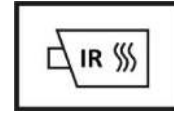
	96V	    
	12V	



	High voltage battery and components		Low voltage battery		High voltage power cable
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USE LARGE AMOUNTS OF PURE WATER



Always utilize full Personal Protective Equipment (PPE) including self-contained breathing apparatus during firefighting operations

Normal firefighting techniques used for conventional ICE vehicles may be used to deal with a fire involving the Ohmio Lift. Making sure to keep applying water to the floor of the vehicle to keep the main batteries cool.

If the main batteries are compromised or catch fire use large amounts of water to extinguish and keep the main batteries cool. Usage of water from a hydrant is highly recommended as other sources of water may run dry before the batteries are sufficiently cool.

Possible reignition of the batteries may occur.

Use Thermal Imaging or Inferred cameras to check for hot spots and cool as necessary.

Small fires may be put out with a Type ABC fire extinguisher.

7. In case of submersion

The Ohmio Lift can be treated as any other submerged vehicle. However, make sure to use appropriate PPE specific to water rescue while handling the vehicle.

Assume that all surfaces of the vehicle are live while it is submerged and until it is completely dried out.

Care should also be taken due to the risk of the main battery catching fire as a result of short circuits created by the water.

8. Towing / Transportation / Storage

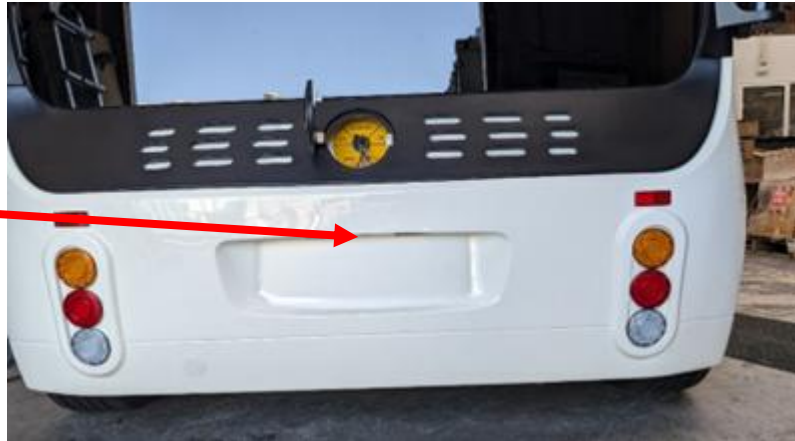
As the Ohmio Lift uses an electric motor, when towing this motor will fight the attempt to be moved. Also, the parking brake will either be engaged or engage after 30 seconds of no speed requests from the robotics system.

Due to these aspects of the vehicle always transport it with all four wheels off the ground using a flatbed towing truck.

MANUAL BRAKE RELEASE

There could be situations where the brake needs to be released, but there is no power to the vehicle, or you need to keep the vehicle off for whatever reason. In those situations, you can release the brake manually by using an external 12V battery connected to a socket located on the switch side of the vehicle, see photo below, and then turning one of the keys located in the switch compartment.

Manual
release



Manual
release



12V battery



RELEASEING THE BRAKE

- Make sure the vehicle is completely switched off
- Connect the battery cable to the vehicle



- Connect the battery cable to the battery, ensuring the polarity is correct (Red to positive, black to negative)



- Turn the key clockwise to disengage the brake
 - **NB:** The key needs to remain in this position until the brake is fully released.



ENGAGING THE BRAKE

- Make sure the vehicle is completely switched off
- Connect the battery cable to the vehicle



- Connect the battery cable to the battery, ensuring the polarity is correct (Red to positive, black to negative)

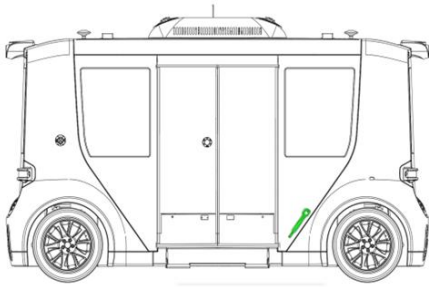


- Turn the key anticlockwise to disengage the brake
 - **NB:** The key needs to remain in this position until the brake is fully engaged

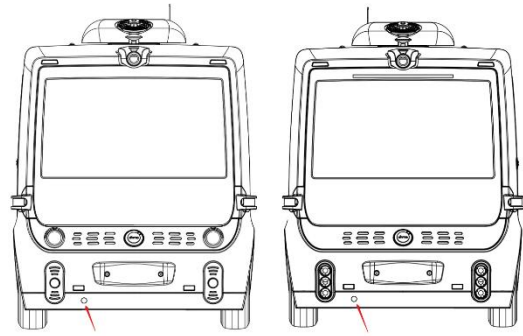


TOW HOOK LOCATION

- Hook located inside interior vehicle compartment



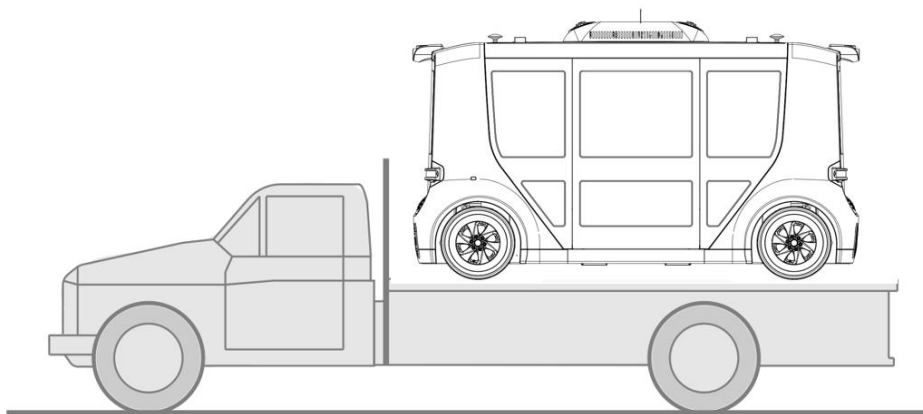
FRONT AND REAR HOOK ATTACHMENT LOCATION



AFTER ANY FIRE OR COLLISION INVOLVING THE BATTERY, STORE VEHICLE OUTSIDE AT LEAST 15M (50FT) FROM OTHER OBJECTS OR VEHICLES

TOWING

- Use a flatbed truck with all wheels off the ground



10. Explanation of pictograms

	Electric vehicle		Li-Ion battery
	Warning high voltage		SLA Battery
	Risk of flammability		Risk of corrosive substance
	Risk of damaging human health		Risk of acute toxicity
	Use water to extinguish the fire		Use IR camera (Thermal imaging)